NIM : 231349015
Nama : Hijriyani Nur Riska

No.
Date:

Latihan Soal - Soal Resistor !

1. Berapakah nilai tahanan untuk resistor berikut :

a) Kuning, Ungu, Emas, Perak

$R = 47 \times 10^{-1} \pm 10\%$

$= 4{,}7\ \Omega \pm 10\%$

b) Orange, Putih, Kuning, Emas

$R = 39 \times 10^{4} \pm 5\%$

$= 390.000\ \Omega \pm 5\%$

$= 390\ k\Omega \pm 5\%$

c) Coklat, Hitam, Merah, Emas

$R = 10 \times 10^{2} \pm 5\%$

$= 1000\ \Omega \pm 5\%$

$= 1k\Omega \pm 5\%$

d) Coklat, Hijau, Hijau, Tidak Berwarna

$R = 15 \times 10^{5} \pm 20\%$

$= 1.500.000\ \Omega \pm 20\%$

$= 1{,}5\ M\Omega \pm 20\%$

e) Hijau, Biru, Hitam, Merah, Coklat

$R = 560 \times 10^{2} \pm 1\%$

$= 56000\ \Omega \pm 1\%$

$= 56k\Omega \pm 1\%$

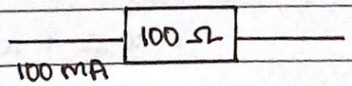
f) Coklat, Abu-abu, Hitam, Hitam, Coklat

$R = 180 \times 10^{0} \pm 1\%$

$= 180\ \Omega \pm 1\%$

2. Hitunglah daya yang didisipasikan pada masing-masing tahanan dengan kondisi tahanan di bawah :

a)



100 mA = 0,1 A

$P = I^2 \times R$

$= (0{,}1)^2 \times 100$

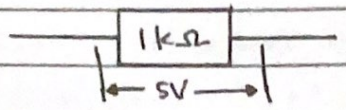
$= 0{,}01 \times 100$

$= 1$ Watt

No.

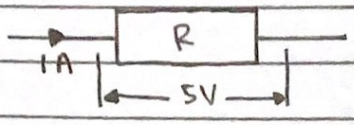
Date:

b)



$P = \frac{V^2}{R} = \frac{(5)^2}{1000} = \frac{25}{1000} = \frac{1}{40} = 0{,}025$ Watt

c)



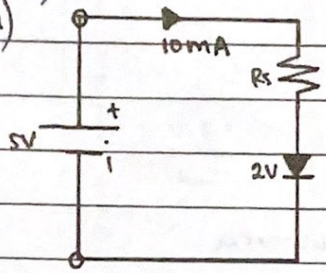
$P = V \times I$

$= 5 \times 1$

$= 5$ Watt

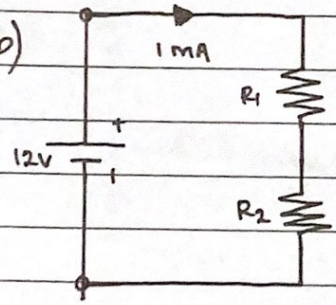
3. Tentukan nilai resistor pada rangkaian di bawah ini yang sesuai dengan nilai standar resistor E_{12}.

a)



$R = \frac{V}{I} = \frac{5}{10 \cdot 10^{-3}} = \frac{5}{10^{-2}} = 5 \times 10^2 = 500\ \Omega$

b)



$R_1 = \frac{V}{I} = \frac{7}{1 \cdot 10^{-3}} = 7 \cdot 10^3 = 7000\ \Omega = 7k\Omega$

$R_2 = \frac{V}{I} = \frac{5}{1 \cdot 10^{-3}} = 5 \cdot 10^3 = 5000\ \Omega = 5k\Omega$

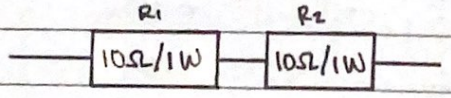
Jadi, $R_{total} = R_1 + R_2$

$= 7K\Omega + 5K\Omega$

$= 12K\Omega$

9. Hitung besar tahanan total dan daya total bila resistor 10 Ω / 1 W disusun secara :

a) Seri



$R_{total} = R_1 + R_2$

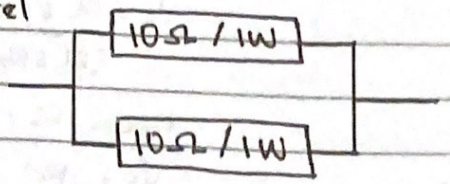
$= 10\ \Omega + 10\ \Omega$

$= 20\ \Omega$

No.

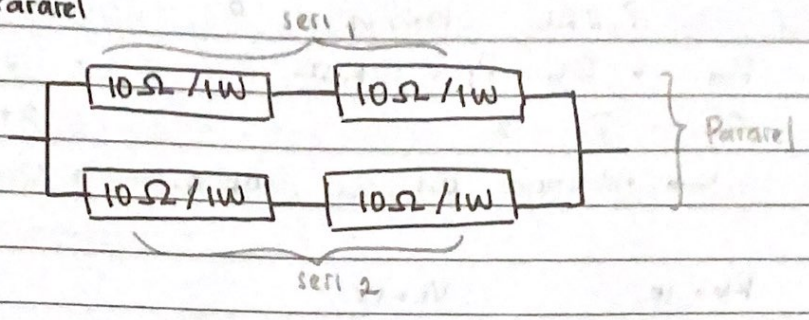
Date:

b) Pararel



$$\frac{1}{R_{total}} = \frac{1}{10} + \frac{1}{10} = \frac{2}{10} = \frac{1}{5} = 5\Omega$$

c) Seri - Pararel



$R_{S1} = 10 + 10 = 20\ \Omega$

$R_{S2} = 10 + 10 = 20\ \Omega$

$$R_P = \frac{1}{R_{S1}} + \frac{1}{R_{S2}}$$

$$= \frac{1}{20} + \frac{1}{20}$$

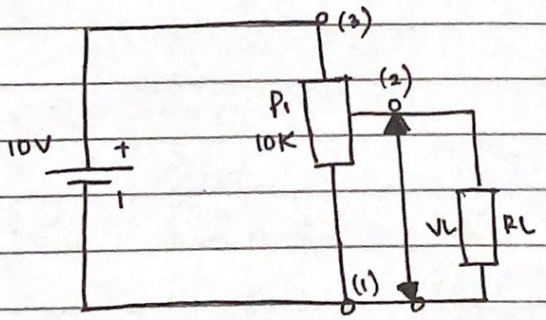
$$= \frac{2}{20}$$

$$= \frac{1}{10}$$

$$= 10\ \Omega$$

5. Gambar Gravik $VL = f(P_{1-2})$ untuk kondisi :

a) $R_1 = \infty$ (tak hingga)



Date:

$V_I = I\,(P \,//\, R_1)$

$= 0\,(0)$

$= 0V$

$I = \dfrac{\mathcal{E}}{P_1 \,//\, RL} = \dfrac{12}{10K/\Omega} = \dfrac{12}{0}$

$Ra \rightarrow Rb = P_1 = 10K\Omega$

$Rb = P_1 - 2$

↳ nilainya $0,1$, ... , $10K\Omega$

$V_L = \dfrac{RL \,//\, Rb}{(RL \,//\, Rb)} \times 10\ V$

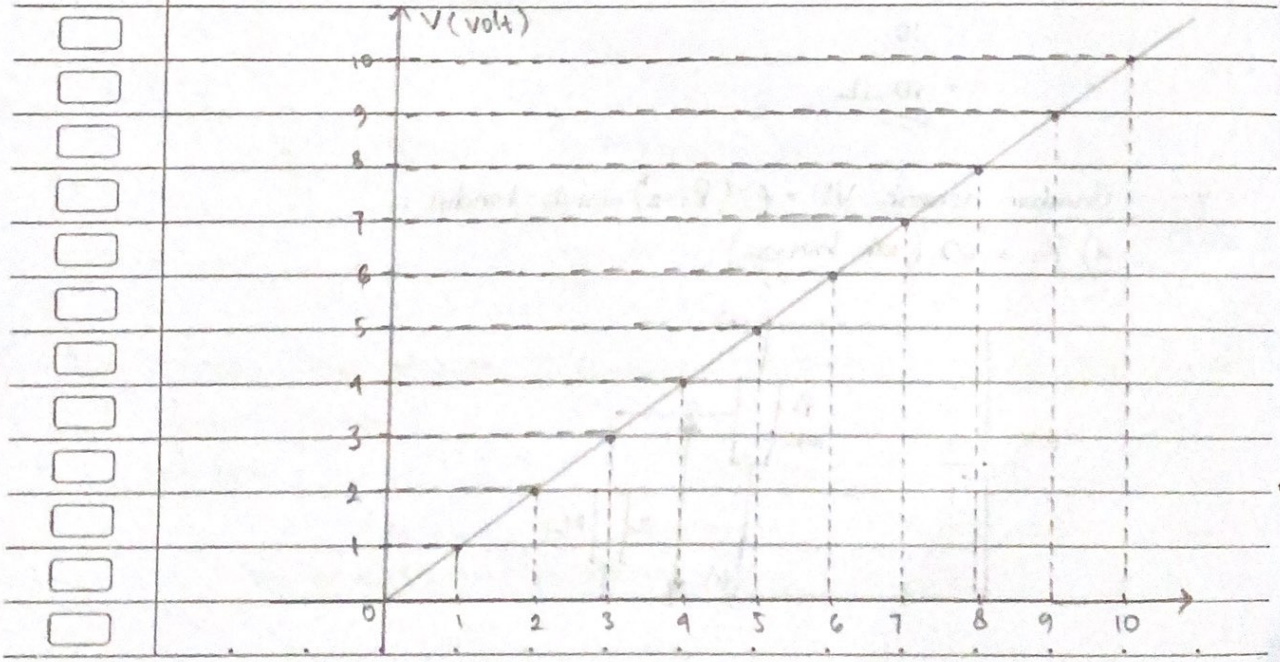
Karena $RL = \infty$ maka,

$V_L = \dfrac{Rb}{Rb + Ra} \times 10$

$= \dfrac{0}{0+10} = 10$

$= 0$

$Rb = 1K$	$V_L = 1V$
$Rb = 2K$	$V_L = 2V$
$Rb = 3K$	$V_L = 3V$
$Rb = 4K$	$V_L = 4V$
$Rb = 5K$	$V_L = 5V$
$Rb = 6K$	$V_L = 6V$
$Rb = 7K$	$V_L = 7V$
$Rb = 8K$	$V_L = 8V$
$Rb = 9K$	$V_L = 9V$
$Rb = 10K$	$V_L = 10V$



No.

Date:

b) $R_L = 5K$

$$V_L = \frac{5K \parallel R_b}{(5K \parallel R_b + R_a)} \times 10$$

$R_b = 0$

$$V_L = \frac{5K \parallel 0}{(5K \parallel 0 + 10)} \times 10 = 0V$$

$R_b = 1K$

$$V_L = \frac{5K \parallel 1K}{(5K \parallel 1K + 9K)} \times 10$$

$R_b = 0$

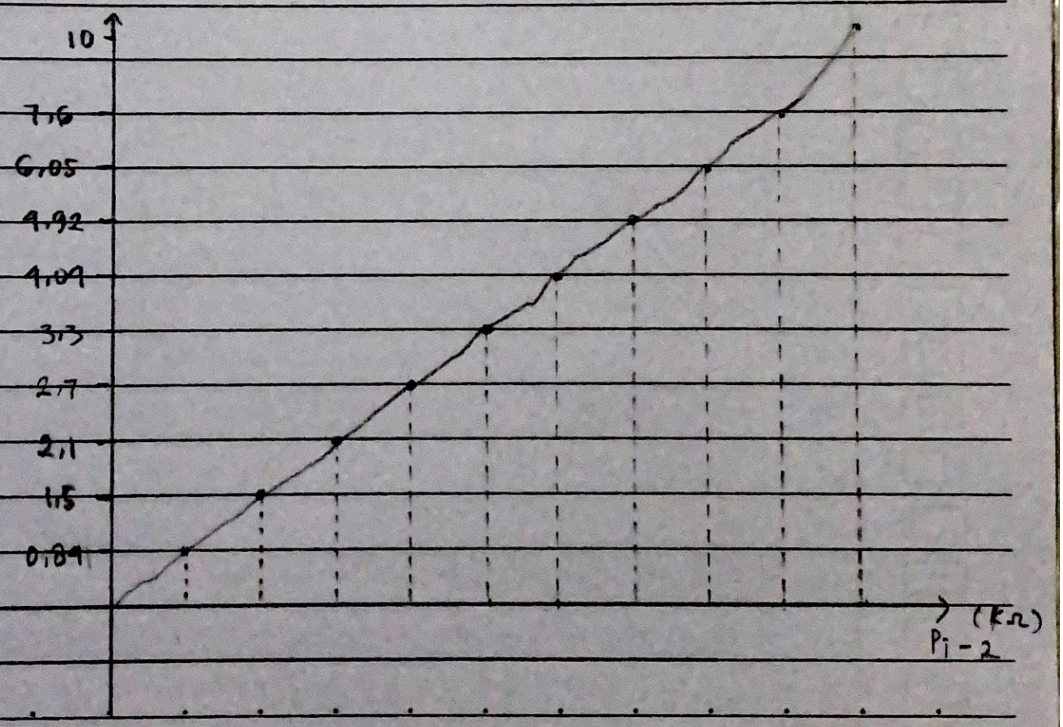
$$V_L = \frac{5k \parallel 0}{(5k \parallel 0 + 10)} \times 10 = 0V$$

$R_b = 1K$

$$V_L = \frac{5K \parallel 1K}{(5K \parallel 1K + 9K)} \times 10 = 0V$$

$R_b = 2k$

$$V_L = \frac{5K \parallel 2K}{(5K \parallel 2K + 8K)} \times 10 = 1.5V$$



No.

Date:

6. Nilai resistor variabel berubah dari 0 - 10 Ω (P_{1-2})

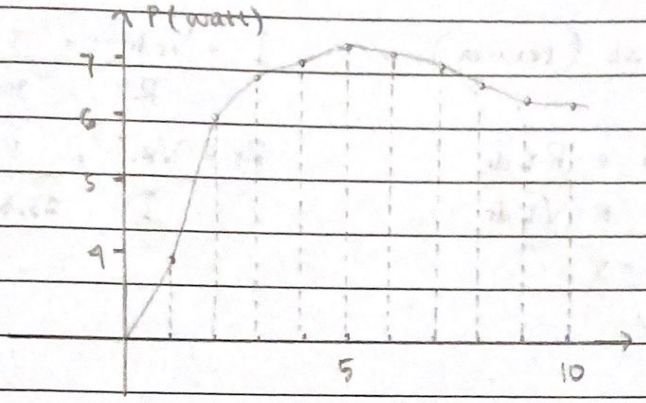
Gambarkan pada diagram yang sama:

a) Grafik daya yang didisipasikan pada Potensiometer P(pot) = f (P_{1-2})

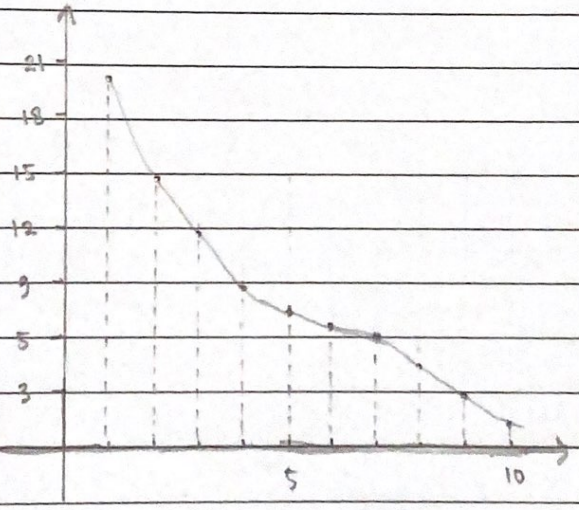
b) Grafik daya yang didisipasikan pada tahanan dalam baterai P(R_i) = f (f [P_{1-2}]

c) Dari hasil grafik yang dibuat pada soal a) dan b), pada kondisi mana terjadi transfer daya maksimum pada beban potensiometer ?

a)



b)



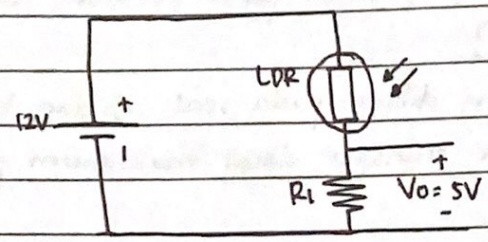
c) Transfer daya maksimum terjadi pada grafik daya yang didisipasikan pada tahanan dalam baterai. Hal tersebut disebabkan karena semakin besar hambatan, daya di potensiometer akan turun.

Date:

7. Tentukan besar nilai R_1 supaya $V_o = 5V$ dalam kondisi terang, bila spesifikasi LDR mempunyai :

Tahanan Gelap (Dark Resistance) = 2 MΩ

Tahanan Terang (Light Resistance) = 300 Ω



R = 300 Ω (terang)

V_o = 5V

$V_s = V_{R1} + R_I dr$

$12 = 5 + V_L dr$

$V_I dr = 12 - 5$

$V_I dr = 7V$

$$I = \frac{V_I dr}{R} = \frac{7}{300} = 23,34 \text{ mA}$$

$$R_1 = \frac{V_{R1}}{I} = \frac{5}{23,34 \cdot 10^{-3}} = 214 \Omega$$